

Module Name & Name of the Module

EBA5005 Practice Module in Specialized Predictive Modelling & Forecasting

Title of Proposal

Proposal for SG HomeFit: A Prescriptive Town & Flat-Type Recommender for Affordability-Constrained Homebuyers

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Team Name or Group No

Group 2

Team Members

Chua Yi Ling — A0315508H

Tan Xue Min Comane — A0204794Y

Daniel Loh Guo Juin — A0149952R

Malur Seshadri Shridhar — A0229909N

1. Introduction

Households planning an HDB resale purchase must balance a fixed budget against requirements such as location, flat size, floor level and remaining lease. These requirements often conflict. Existing tools, such as transaction listings and affordability calculators, tell a household what it can afford. They do not show which compromises give the most value, or what relaxing a requirement would be worth. Resale prices also move, which adds uncertainty for households planning to buy in the next 6–12 months.

This project proposes a predictive-to-prescriptive decision-support prototype for HDB resale housing. It will (i) forecast near-term resale prices for indicative housing profiles, (ii) estimate how households value competing attributes, and (iii) use constrained optimisation to return a ranked shortlist of profiles that fit a household's budget and essential requirements. It will also show how the shortlist changes as the budget or requirements are relaxed, making the trade-offs explicit.

2. Industry Overview

HDB resale housing provides a relevant setting for studying housing choice under budget constraints. Resale flats differ in location, flat type, floor area and remaining lease, creating trade-offs for households with different needs. Historical transaction records provide a basis for examining price differences and trends, although they do not represent current property listings.

The project will focus on selected HDB resale towns and flat types. It will use a household's stated purchase budget and housing requirements as inputs. Financing approval, grants and statutory eligibility assessment will remain outside the prototype's scope.

3. Business Problem & Objectives

Households have a largely fixed budget and several goals that often conflict: affordability, proximity to MRT and schools, floor level and adequate remaining lease. Existing tools report what a household can afford, but do not rank options against these goals, or show what relaxing one goal would gain in another. Near-term price movements and the limited supply of matching flats add further uncertainty.

Business Objectives:

- For a household's stated budget, minimum requirements and preferences, return a short, ranked and characteristically diverse list of housing profiles that stay within budget across the forecast price range.
- Quantify the "price" of each requirement: how much the best option improves with extra budget or a shorter minimum lease, and what insisting on a longer lease cost.
- Indicate how readily each shortlisted profile comes up for sale, so households can gauge how long a search may take.
- Demonstrate a decision-support approach usable by proptech platforms or housing advisory services, beyond a one-off price forecast.

Technical Objectives:

- Pricing and Forecasting: price profiles with a transaction-level hedonic model and forecast them 6-12 months ahead with SARIMAX, benchmarked against naive baselines and a pooled LSTM. Use SHAP to confirm the drivers are economically sensible.
- Preferences: estimate part-worth utilities, willingness-to-pay and price elasticity of choice through choice-based conjoint.
- Availability: model the time until a matching flat transacts, using Kaplan-Meier and Cox proportional hazards.
- Prescription: select the shortlist with mathematical and goal programming, and value each constraint through sensitivity analysis.

4. Project Design

- **Data and features:** Acquire HDB resale transactions and block attributes. Geocode blocks and compute remaining lease, distance to MRT, nearby primary schools, availability of supermarkets, hawker centres, parks, healthcare facilities and storey band. Define indicative housing profiles.
- **Pricing:** Fit a transaction-level hedonic price model and use it to derive a quality-adjusted monthly price index by region and flat type. Check the model's drivers with SHAP.
- **Forecasting:** Forecast the index 6–12 months ahead with SARIMAX, benchmarked against naive baselines and a pooled LSTM. Apply the forecasts to profile prices, with prediction intervals.
- **Availability:** Build transaction-gap histories by block and flat type. Fit Kaplan–Meier curves and a Cox proportional hazards model to estimate the expected wait for a matching flat in each profile.
- **Preferences:** Design and field a choice-based conjoint survey, subject to any required ethics clearance. Estimate part-worth utilities, willingness-to-pay (WTP) and price elasticity of choice.
- **Prescription:** Formulate the shortlist as a mathematical and goal programme. Solve it for representative household profiles and run budget and lease sensitivity sweeps.
- **Validation and delivery:** Back-test the recommender, build the what-if dashboard, and consolidate into the report.

5. Scope of work

Datasets to be Used:

- HDB Resale Flat Prices (Jan 2017-present), data.gov.sg.
 - Transaction-level records: month, town, flat type, block, storey range, floor area, flat model, lease commencement, price.
- HDB Property Information, data.gov.sg.
 - Year completed and number of units by flat type for each block. This supplies the survival model's exposure and the MOP-supply regressor.
- Locational. SLA OneMap API (block geocoding), LTA DataMall (MRT/LRT station locations), MOE School Directory (primary schools), data.gov.sg (Amenities & Facilities such as Hawker Centres, Supermarket Licences, Parks, CHAS Clinics; Polyclinics, Childcare & Kindergarten Centres).
- Macro regressors such as interest-rate series (MAS), dates of property cooling measures, and flats reaching MOP (from completion year).
- Preference data, via a choice-based conjoint survey.
 - Attributes are price, distance to MRT, remaining lease, storey and school proximity. Each respondent sees about 8 choice tasks of 3 alternatives, plus one hold-out task. The target is at least 50 respondents from a convenience sample, which is noted as a limitation.

Proposed Models, Techniques, and Evaluation:

Stage	Technique	Role	Evaluation
Predict	Hedonic price model	Prices each profile today. Month effects give a quality-adjusted price index	Hold-out error on recent transactions
	XAI/SHAP (ACE)	Explains the hedonic model's price drivers, and later the LSTM benchmark	Drivers economically plausible (lease, MRT, storey, size)
	SARIMAX (CPMF)	Forecasts the index 6–12 months ahead by region and flat type. Regressors: interest rate, MOP supply, cooling-measure dummies. Output: profile price with interval.	Walk-forward MAPE vs naive and seasonal-naive baselines; interval coverage
	LSTM (CPMF)	Benchmark forecaster, pooled across all series	Diebold–Mariano test
	Kaplan Meier, Cox Proportional Hazard Model	Time until a matching flat next transacts, by block and flat type. Output: expected wait per profile.	C-index on a later period; proportional-hazards check

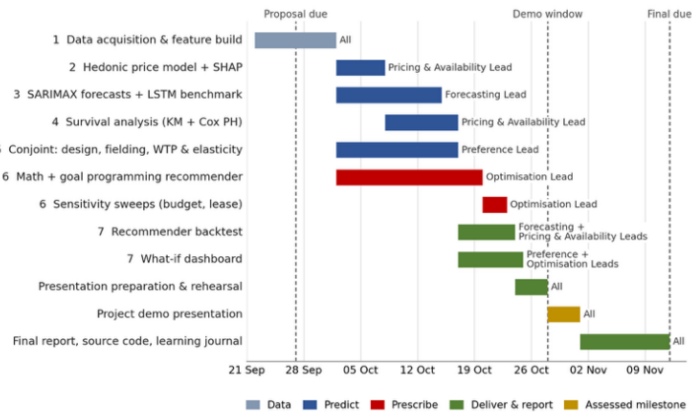
	(CPMF)		
	Conjoint Analysis (CPMF)	Part-worth utilities from a choice-based survey, used as default preference weights	Hold-out task hit rate
	Price Elasticity (PPA)	Willingness-to-pay (WTP) and price elasticity of choice from the conjoint price coefficient. WTP expresses every goal in dollars.	Plausible signs and magnitudes, with confidence intervals
Prescribe	Mathematical Programming (PPA)	Selects a diverse shortlist of k profiles that maximises WTP-valued surplus. Hard constraints: budget $\geq$ upper forecast price, minimum lease, at most one profile per town.	Feasibility and constraint checks on representative households
	Goal Programming (ACE)	Soft goals (MRT, school, amenities, facilities, storey, expected wait) as dollar-weighted deviations from household targets	Shortlists change sensibly as preferences are varied
	Monte Carlo Simulation	Simulates buy-now vs wait-6/12-months outcomes for each shortlisted profile. It draws price paths from SARIMAX under interest-rate and supply scenarios, and combines them with survival-based availability.	Calibration: in the backtest, how often do realised prices fall within the simulated ranges?
	Sensitivity/Shadow Price (PPA)	Parametric sweeps of budget and minimum lease give the marginal value of each constraint	Monotonicity (more budget never makes the result worse)
Deliver	Recommender backtest	Generates shortlists at past dates using only the data available then	Share of shortlisted profiles whose realised price 6–12 months later stayed within budget, vs a naive "cheapest towns" baseline
	What-If Dashboard	Households enter budget and preferences and see the shortlist, expected wait and constraint trade-offs	Walk-through with representative household scenarios

6. Key Deliverables

- Cleaned, feature-engineered transaction dataset and the housing-profile definitions
- Predictive Analysis:
 - A hedonic price model with a SHAP driver analysis
 - SARIMAX price forecasts 6–12 months ahead by region and flat type, compared against LSTM and naive benchmarks
 - Kaplan–Meier curves and Cox hazard ratios, giving the expected wait per profile
 - The conjoint survey instrument, part-worth utilities, willingness-to-pay and price-elasticity estimates
- Prescriptive Analysis: a mathematical and goal programming recommender that returns a diverse, ranked shortlist, with budget and lease sensitivity analysis
- Backtest results against a naive baseline, and a what-if dashboard
- Final group report, source code, presentation slides

7. Effort Estimates and Timeline

Gantt chart (team of 4, ~10 person-days each; dates are calendar activity windows):



#	Activity	Owner	Start	End	Effort (MDs)
1	Data acquisition & feature build	All	22 Sep 2026	1 Oct 2026	6
2	Hedonic price model + SHAP	Pricing & Availability Lead	2 Oct 2026	7 Oct 2026	2.5
3	SARIMAX forecasts + LSTM benchmark	Forecasting Lead	2 Oct 2026	15 Oct 2026	5
4	Survival analysis (KM + Cox PH)	Pricing & Availability Lead	8 Oct 2026	16 Oct 2026	3
5	Conjoint: design, fielding, WTP & elasticity	Preference Lead	2 Oct 2026	16 Oct 2026	5
6	Math + goal programming recommender	Optimisation Lead	2 Oct 2026	19 Oct 2026	4
6	Sensitivity sweeps (budget, lease)	Optimisation Lead	20 Oct 2026	22 Oct 2026	1.5
7	Recommender backtest	Forecasting + Pricing & Availability Leads	17 Oct 2026	23 Oct 2026	2.5
7	What-if dashboard	Preference + Optimisation Leads	17 Oct 2026	24 Oct 2026	2.5
-	Presentation preparation & rehearsal	All	24 Oct 2026	27 Oct 2026	4
-	Project demo presentation	All	28 Oct 2026	31 Oct 2026	—
-	Final report, source code, learning journal	All	1 Nov 2026	11 Nov 2026	6

8. References

- HDB Resale Flat Prices (from Jan 2017) — data.gov.sg, https://data.gov.sg/datasets/d_8b84c4ee58e3cfc0ece0d773c8ca6abc/view
- HDB Renting Out of Flats (from Jan 2021) — data.gov.sg, <https://data.gov.sg/collections/189/view>
- HDB eligibility rules — income ceiling and Minimum Occupation Period — HDB.gov.sg
- OneMap API — SLA, <https://www.onemap.gov.sg/apidocs/>; Static Datasets — LTA Datamall, <https://datamall.lta.gov.sg/>
- Table Builder (household income distribution, CPI) — SingStat, <https://tablebuilder.singstat.gov.sg/>
- School Directory and Information — MOE/data.gov.sg